

Which Genes Leak? Feature-Level Privacy Auditing for Synthetic Single-Cell RNA-Seq Data

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Abstract—Synthetic single-cell RNA-seq (scRNA-seq) data is shared as a privacy-preserving alternative to raw genomics, yet recent membership inference attacks (MIA) show that generators memorize training donors. Current auditing detects that leakage occurs but not where in the feature space it originates, leaving custodians with only blanket differential privacy as a defense. We introduce Gene-Level Privacy Auditing (GLPA), which attributes MIA success to individual genes via SHAP analysis of membership-inference attack scores. Across three generators (GMM, VAE, GAN) and both real (PBMC 3k) and simulated datasets, GLPA reveals that privacy leakage is highly concentrated: the top 5% of genes account for 66 to 73% of the total membership-inference signal (Gini > 0.90). Leakage patterns are also subgroup-dependent, with different donor groups leaking through largely distinct gene sets. These findings suggest that privacy risk in synthetic genomic data is localized rather than uniformly distributed across features. Targeted perturbation of just 50 genes reduces MIA AUC by up to 26 points while preserving $>90\%$ of differential expression structure, and is more effective than random-gene perturbation at a matched budget (up to $3.18\times$ at the generator-specific optimum). GLPA gives custodians an actionable pre-release tool that pinpoints which features and subgroups require protection.

Index Terms—privacy auditing, synthetic data, single-cell RNA-seq, membership inference attack

I. INTRODUCTION

Single-cell RNA sequencing (scRNA-seq) has revolutionized our understanding of cellular heterogeneity, disease mechanisms, and developmental biology. However, the sensitive nature of genomic data necessitates strict access controls, creating barriers to the open science that drives discovery. Synthetic data generation has emerged as a promising solution: train a generative model on real donor data, release the synthetic output, and retain the privacy of original donors.

This premise is increasingly under scrutiny. Recent membership inference attacks (MIA) demonstrate that synthetic scRNA-seq data can reveal whether a specific donor was used to train the generator [1], [2], particularly in the small-cohort settings common in biomedical studies. While existing auditing methods can detect the presence of privacy leakage, they provide little insight into which genes drive the attack signal. As a result, data custodians are often limited to

blanket defenses such as differential privacy (DP), which treat all genes as equally sensitive and can substantially degrade biological utility. Understanding where leakage originates in the feature space is therefore essential for designing more targeted and utility-preserving protections.

We address this gap with **Gene-Level Privacy Auditing (GLPA)**. Our contributions are:

- 1) A SHAP-based attribution method that ranks individual genes by their contribution to membership-inference scores, producing a per-generator “leaky gene” list.
- 2) Empirical evidence that leakage is highly concentrated and subgroup-specific: control and disease donors leak through nearly disjoint gene sets (Jaccard 0.02 to 0.10), so a universal defense is misallocated.
- 3) A characterization of the biology of leaky genes: they are largely distinct from disease markers, cleanly so in simulation and with only a small real-data overlap (§VI-I), which bounds the utility cost of targeted perturbation.
- 4) A targeted perturbation scheme that beats budget-matched uniform noise, validated causally against random-gene baselines.

II. RELATED WORK

Membership inference attacks. Since Shokri et al. [3] introduced shadow-model MIA, a large body of work has made the attack cheaper and stronger: ML-Leaks relaxes the shadow-model and same-distribution data assumptions [4], white-box analyses exploit gradients and intermediate activations [5], and Bayes-optimal analyses show the per-example loss is a near-sufficient statistic for membership [6], which LiRA [7] operationalizes as a calibrated per-example likelihood-ratio test. The common thread is a tight connection to overfitting and memorization [8] (see [9] for a survey). MIA applies directly to generative models, where attacks exploit the proximity of synthetic samples to memorized training points [10]–[12], and to synthetic releases across modalities: tabular [13], [14], health records [15], general synthetic-training pipelines [2], and synthetic scRNA-seq [1], with CAMDA’25 [16] formalizing the last as a benchmark. We extend this line by asking which features drive the leakage, not merely whether it occurs.

Disparate vulnerability and feature attribution. MIA risk varies across subgroups [17], [18], and fairness interventions can amplify the gap [19]. We lift this analysis from attack

The source code is available at <https://github.com/ShuyaFeng/-Auditing-for-Synthetic-Single-Cell-RNA>.

rates to the actual feature sets that leak. SHAP [20] is widely used for model explanation, and SHAP-based regularization has recently been used to reduce leakage during training [21]. We instead apply SHAP post-hoc to audit which features of the data enable membership inference.

Genomic privacy and differential privacy. Genomes are uniquely identifying: individuals can be re-identified from summary statistics [22], by surname inference against public databases [23], or through data-sharing beacons [24], motivating a broad literature on genetic-privacy threats and countermeasures [25]. Differential privacy (DP) [26]–[28] is the dominant formal defense, bounding the influence of any single record on a released computation. For deep models it is realized through differentially private optimization [29], [30], maturing into DP-SGD [31] with tight privacy accounting via Rényi DP [32] and ensemble alternatives such as PATE [33]. DP has also been applied to genomic data dissemination [34], while selective-hiding methods share only non-sensitive SNPs [35]. These defenses act uniformly or during training. Our targeted perturbation instead operates post-hoc on synthetic outputs with data-driven, generator-specific identification of sensitive features.

Generative models for single-cell data. Synthetic scRNA-seq is produced both by statistical simulators with explicit zero-inflated count models [36] and by deep generative models, including variational autoencoders [37], [38], generative adversarial networks [39], and, more recently, diffusion models [40]. We audit representative statistical and deep generators (GMM, VAE, GAN) rather than proposing a new one. GLPA is agnostic to the generator family and treats each as a black box that emits synthetic profiles.

III. THREAT MODEL AND PROBLEM FORMULATION

A. Setting

Consider a data custodian holding scRNA-seq data from N donors, with donor-level expression profiles $\mathbf{X} = \{x_1, \dots, x_N\} \subset \mathbb{R}^G$ where G is the number of genes. The custodian trains a generative model $\mathcal{G}$ on a subset $\mathcal{M} \subset \mathbf{X}$ (the members) and releases synthetic data $\hat{\mathbf{X}} = \mathcal{G}(\mathcal{M})$.

Adversary. An adversary with access to $\hat{\mathbf{X}}$ and a target record x_t aims to determine whether $x_t \in \mathcal{M}$. The adversary may also have access to auxiliary data from the same distribution to train shadow models.

Auditor. The data custodian acts as a privacy auditor before release. Unlike the adversary, the auditor has full access to $\mathcal{G}$, $\mathcal{M}$, and the non-member set. The auditor’s goal is to identify which genes $g \in \{1, \dots, G\}$ contribute most to MIA success, enabling targeted mitigation.

B. Gene-Level Leakage Score

We define the **leakage attribution** of gene g as:

$$\phi_g = \frac{1}{N} \sum_{i=1}^N |\text{SHAP}_g(x_i, f)| \quad (1)$$

where $f: \mathbb{R}^G \rightarrow \mathbb{R}$ is a model trained to predict membership score from gene expression, and $\text{SHAP}_g(x_i, f)$ is the SHAP

value of gene g for record x_i . Genes are ranked by ϕ_g in descending order to produce the leaky gene ranking π .

C. Targeted Perturbation Objective

Given the leaky gene ranking π , the targeted perturbation problem is:

$$\min_{k, \sigma} \text{MIA-AUC}(\hat{\mathbf{X}}^{(k, \sigma)}) \quad \text{s.t.} \quad \text{Utility}(\hat{\mathbf{X}}^{(k, \sigma)}) \geq \tau \quad (2)$$

where $\hat{\mathbf{X}}^{(k, \sigma)}$ is the synthetic data after adding noise of scale σ to the top- k genes in ranking π , and τ is the minimum acceptable utility (e.g., preserving at least 90% of differential-expression structure).

IV. METHOD

The GLPA pipeline has four stages: (1) train a synthetic generator on the member donors, (2) run a LiRA-style membership inference attack to obtain per-donor leakage scores, (3) attribute those scores to individual genes via SHAP, and (4) apply targeted perturbation to the top-ranked leaky genes and re-evaluate. We describe each stage below.

A. Membership Inference via LiRA

We adapt the Likelihood Ratio Attack (LiRA) [7] for the synthetic data setting:

- 1) **Shadow model training.** Train $S = 16$ shadow generators, each on a random half of all donors. Record which donors are “in” vs. “out” for each shadow.
- 2) **Membership scoring.** For each donor x_i and each shadow model s , compute a proximity score $c_s(x_i) = -\frac{1}{k_{\text{nn}}} \sum_{j=1}^{k_{\text{nn}}} \|x_i - \text{NN}_j^{(s)}\|$ as the mean distance to its k_{nn} nearest neighbors in the shadow’s synthetic data (the neighbor count k_{nn} is distinct from the top- k perturbation budget used later).
- 3) **Likelihood ratio.** For each donor, fit Gaussians $\mathcal{N}(\mu_{\text{in}}, \sigma_{\text{in}}^2)$ and $\mathcal{N}(\mu_{\text{out}}, \sigma_{\text{out}}^2)$ to the in-model and out-model score distributions. The LiRA score is the log-likelihood ratio.
- 4) **Direct scoring.** Additionally compute the k_{nn} -NN distance to the target generator’s synthetic data as a complementary signal.

B. Gene-Level Leakage Attribution

Given per-donor membership scores from LiRA, we attribute leakage to individual genes:

- 1) **Attribution model.** Train a gradient-boosted tree (XGBoost) to predict the combined membership score from donor-level gene expression. This score is the per-donor aggregate of the two signals defined above, the LiRA log-likelihood ratio and the direct k_{nn} -NN proximity, and at our sample size it is dominated by the informative direct component (§VI-A). The model captures non-linear gene interactions that drive memorization.
- 2) **SHAP computation.** Apply TreeSHAP [41] to compute per-gene, per-donor SHAP values. The mean absolute SHAP value per gene yields the leakage attribution ϕ_g .

- 3) **Stability analysis.** Bootstrap the attribution pipeline (resample donors, retrain XGBoost, recompute SHAP) $B = 10$ times. Report Jaccard stability of the top- k gene sets across bootstrap iterations.
- 4) **Subgroup stratification.** Run the full attribution pipeline separately for each donor subgroup (e.g., disease vs. control). Compare resulting gene rankings to quantify subgroup-specific leakage.

C. Biological Characterization

Cross-reference the top- k leaky genes with known biological categories:

- **Disease-associated genes:** Genes with strong differential expression between subgroups.
- **High-variance genes:** Genes with large inter-donor expression variability.
- **Normal genes:** The remaining genes.

Compute enrichment (fold over random expectation) and statistical significance (Fisher’s exact test) for each category.

D. Targeted Perturbation Mechanisms

We implement four targeted perturbation strategies, each applied only to the top- k leaky genes: (1) **Selective Laplace noise** adds $\text{Lap}(0, \sigma \cdot \text{std}(g))$ to each leaky gene g , scaled by its standard deviation. (2) **Gene-level quantization** discretizes expression values of leaky genes into $B = 10$ equal-frequency bins, replacing each value with its bin center. (3) **Subgroup-conditional smoothing** replaces individual expression values with subgroup means for leaky genes only. (4) **Budget-matched uniform noise (baseline)** distributes the same total noise budget uniformly across all G genes: $\sigma_{\text{uniform}} = \sigma_{\text{selective}} \cdot k/G$. Throughout, “budget” denotes the total injected noise magnitude (the sum of per-gene σ values across perturbed genes), not a differential privacy budget ϵ . Budget-matched comparisons use identical total noise and differ only in how it is distributed across genes.

We emphasize that these are heuristic post-hoc mitigations rather than formal (ϵ, δ) -DP mechanisms: noise is added to synthetic outputs (not to a query of real data), scaled by per-gene standard deviation rather than worst-case sensitivity, and applied to a top- k set that is itself selected from the audited data. We therefore evaluate them by measured MIA reduction rather than a worst-case privacy budget, and treat them as complementary to, not a replacement for, formally guaranteed in-training defenses such as DP-SGD [31] or PATE [33].

V. EXPERIMENTAL SETUP

Datasets. We evaluate on two datasets. The first is PBMC 3k [42], processed with Scanpy [43] into 60 pseudo-donors split into 30 myeloid/B-cell-enriched and 30 T/NK-enriched subgroups, retaining the top 978 variable genes. The second is a controlled simulation of $N=60$ donors and $G=978$ genes using zero-inflated negative binomial sampling [36] with latent-factor gene-gene correlations, donor-specific signatures, and subgroup effects on 30 designated disease genes, giving known disease-marker and subgroup structure against

TABLE I
MEMBERSHIP INFERENCE RESULTS. DIRECT k -NN SCORING RE-IDENTIFIES MEMBERS ACROSS ALL THREE GENERATORS, OUTPERFORMING LiRA AT THIS SAMPLE SIZE.

Generator	Direct MIA AUC	LiRA AUC	TPR@1%FPR
GMM	0.994	0.392	0.033
VAE	0.978	0.348	0.000
GAN	0.787	0.520	0.167

which to interpret leaky-gene composition. The 978-gene budget matches the L1000 landmark panel [44], keeping both datasets comparable. On the simulation we also sweep $N \in \{20, 40, 60, 100, 200\}$ to study scaling. For both, expression is log-normalized and clipped to $[0, 15]$, and we fix a random member/non-member split with $n_{\text{mem}} = n_{\text{non}} = 30$ donors, stratified by subgroup, so all generators and attacks see identical training and evaluation populations.

Generators. We use three generators that span statistical and deep generative families and are all trained on the same 30 member donors. The GMM uses 5 mixture components with full covariance and serves as a strong memorization baseline. The VAE [37] is a fully-connected encoder-decoder with 256 to 512 hidden units and a latent space of 32 to 64 dimensions, trained for 200 to 300 epochs with KL annealing. The GAN [39] uses 256 to 512 hidden units and a latent space of 64 to 128 dimensions, trained for 300 to 500 epochs with a standard non-saturating loss. Each generator emits 30 synthetic donor profiles for evaluation.

Evaluation metrics. For privacy we report MIA AUC, where 0.5 is chance and 1.0 is perfect re-identification, together with TPR at 1% FPR. Following Carlini et al. [7], TPR at low FPR captures the worst-case threat of a confident attack that AUC can hide. With only $n = 30$ non-members it is necessarily coarse, resolved to multiples of $1/30$, so we report it for comparability with prior work rather than as a precise operating point. For utility we report differential expression (DE) preservation, measured as the Spearman correlation of between-subgroup fold changes, random-forest subgroup classification accuracy, gene-gene correlation preservation under the normalized Frobenius distance, and clustering ARI and NMI against ground-truth labels. Together these cover the supervised, unsupervised, and co-expression-structure analyses that downstream users care about.

VI. RESULTS

A. Membership Inference Attacks Succeed Across Generators

All three generators are vulnerable to MIA using direct k -NN scoring against the synthetic data (Table I). The GMM, which directly models the training distribution, is most vulnerable, the VAE nearly as much, and the GAN moderately but non-trivially so (Table I). All three generator classes therefore memorize training data sufficiently for MIA, consistent with prior findings on synthetic data privacy [1], [2].

The LiRA AUCs sit at or below chance (0.35 to 0.52). With only $n = 30$ members and $S = 16$ shadow generators, the

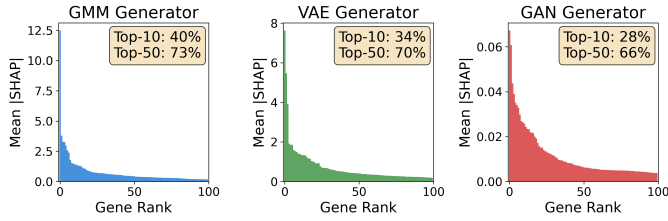


Fig. 1. Leakage concentration across generators. SHAP attribution shows that a small minority of genes carries the bulk of the membership signal. Cumulative SHAP curves are heavily skewed, and Gini coefficients exceed 0.90 for all three generators.

TABLE II

LEAKAGE CONCENTRATION STATISTICS. THE TOP 50 GENES (5% OF FEATURES) CARRY 66 TO 73% OF TOTAL SHAP ATTRIBUTION, AND GINI EXCEEDS 0.90 FOR EVERY GENERATOR.

Generator	Top-10 %	Top-50 %	Top-100 %	Gini
GMM	39.6%	73.1%	89.0%	0.923
VAE	34.2%	70.2%	87.0%	0.915
GAN	28.3%	65.8%	83.3%	0.900

per-donor in/out score distributions are too sparse to calibrate the likelihood ratio reliably. Because LiRA’s power is known to require far more shadow models [7], the shadow-calibrated score is uninformative and, when miscalibrated, can fall below chance. We therefore use the direct k_{nn} -NN proximity to the target generator’s output as the membership signal and base the gene-level attribution on it, retaining LiRA only as a complementary input to the combined score.

B. Leakage Concentrates in a Small Gene Subset

SHAP attribution shows that leakage is highly concentrated (Figure 1). For the GMM generator, the top-ranked gene (Gene_93) reaches a mean absolute SHAP value of 12.49 in predicted-membership-score units, orders of magnitude above the median gene, whose contribution is negligible. We quantify this concentration with two measures: cumulative SHAP fractions and Gini coefficients.

Gini coefficients above 0.90 indicate extreme inequality: a tiny minority of genes carries the vast majority of the leakage signal. The top 50 genes (5% of features) account for 66 to 73% of total leakage attribution across all generators. This concentration has a direct practical implication: privacy auditing need not consider all genes equally. A small, identifiable subset carries the privacy risk.

C. Subgroup Leakage Profiles Are Strikingly Different

Our most striking finding is the divergence of leakage profiles across donor subgroups (Figure 2).

The two subgroups leak through almost entirely different genes: only 4/50 (GMM), 5/50 (VAE), and 2/50 (GAN) of the top-50 leaky genes are shared, a Jaccard similarity of just 0.042, 0.053, and 0.020, respectively (Table VI). This has three implications. First, one-size-fits-all defenses are suboptimal, since a strategy tuned to the aggregate population misses subgroup-specific channels. Second, privacy auditing must

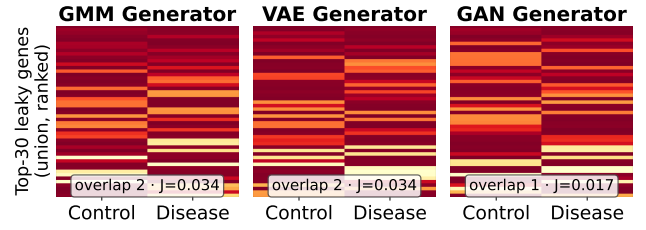


Fig. 2. Subgroup leakage heatmap (simulated data). Top-50 leaky genes for control vs. disease subgroups are nearly disjoint (Jaccard 0.02 to 0.05, with the wider 0.02 to 0.10 range quoted elsewhere also including PBMC), revealing that each subgroup leaks through a distinct gene set.

be subgroup-aware, because the genes enabling MIA against disease and control donors differ. Third, the divergent channels are not the engineered disease-marker genes (§VI-D), so they cannot be read off from the disease signal alone.

D. Leaky Genes Are Not Disease-Associated (Simulation)

A key question for deployment: do leaky genes overlap with biologically important disease markers? If they do, suppressing them would destroy the synthetic data’s primary utility. On the controlled simulation, we test this with Fisher’s exact test for enrichment of disease-associated and high-variance genes among the top-50 leaky genes (Table VII and Figure 4). The corresponding real-PBMC analysis, with gene names and curated immune pathways, appears in §VI-I.

Neither disease genes nor high-variance genes are significantly enriched among the leaky gene sets (all $p \geq 0.46$). With only 30 disease genes and a 50-gene set (expected overlap ≈ 1.5), this test has limited power, so the result is best read as no detectable enrichment rather than proven independence. On the simulation, then, **leaky genes can largely be suppressed without disrupting the engineered subgroup signal**: the features that enable membership inference are distinct from the designated disease genes. We caution that this clean separation is a property of the controlled simulation. On real PBMC data we find a narrower but real overlap with subgroup-defining markers (§VI-I), which the deployment workflow in §A accounts for.

For the GMM, slight enrichment of high-variance genes (1.17 $\times$) suggests that individual expression variability contributes to memorization, but not at a statistically significant level. The GAN shows zero overlap with disease genes, indicating that its leakage arises from artifacts of the generative process rather than biologically meaningful patterns.

E. Targeted Perturbation Outperforms Uniform Noise

We compare selective perturbation, which adds noise to the top- k leaky genes only, against budget-matched uniform noise, which spreads the same total noise budget across all genes. Figure 3 shows the privacy-utility frontier, and Table III reports the GAN and VAE numbers at $\sigma = 2.0$.

For the GAN, selective perturbation of 50 genes reduces MIA AUC by 0.035 versus 0.013 for uniform noise, a **2.7 $\times$ greater privacy improvement** at the same budget, while

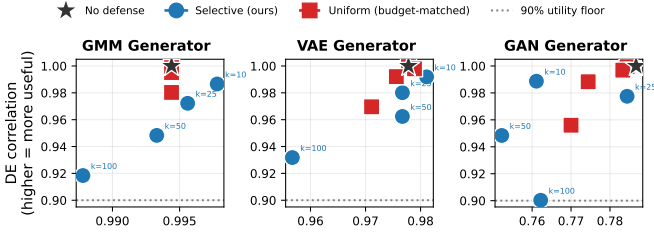


Fig. 3. Privacy-utility Pareto frontier across generators. The horizontal axis is MIA AUC (lower = more private), and the vertical axis is DE correlation (higher = more useful). Selective (top- k) perturbation dominates budget-matched uniform noise: at the same DE-preservation budget, targeted perturbation reduces MIA AUC further.

TABLE III
PERTURBATION RESULTS AT $\sigma = 2.0$. SELECTIVE PERTURBATION REDUCES MIA AUC 2.7 TO 3.0 $\times$ MORE THAN UNIFORM NOISE AT THE SAME BUDGET WHILE PRESERVING $\geq 93\%$ OF DE STRUCTURE.

Gen.	Method	k	MIA AUC	Δ AUC	DE Corr.
GAN	None	—	0.787	—	1.000
GAN	Selective	50	0.752	−0.035	0.948
GAN	Uniform	50	0.774	−0.013	0.988
VAE	None	—	0.978	—	1.000
VAE	Selective	100	0.957	−0.021	0.932
VAE	Uniform	100	0.971	−0.007	0.970

preserving 94.8% of DE correlation structure. The VAE shows the same pattern at $k = 100$ (3.0 $\times$). Moderate post-hoc noise is insufficient for the GMM (AUC 0.994), where privacy protection must instead be incorporated during generation, for example via DP-SGD. We treat this as a useful diagnostic for practitioners choosing architectures.

High-noise regime. Scaling the noise to $\sigma = 8.0$ on the same top-50 leaky genes (5% of features) drops GMM MIA AUC from 0.994 to **0.732** (−0.262), VAE from 0.978 to 0.838 (−0.140), and GAN from 0.787 to 0.667 (−0.120), while DE preservation stays above 90% in every case. Targeted perturbation of a small gene subset therefore achieves meaningful privacy protection even for highly memorizing generators, leaving 95% of the feature space untouched. Because perturbation touches only the top- k genes, some residual membership signal necessarily remains in the untouched tail. The steep concentration curves (Table II) bound how large this residual can be, and it shrinks further as k and σ increase, recovering the privacy-utility trade-off a custodian can dial to taste. The full $k \times \sigma$ ablation is reported in Appendix B (Figure 5).

F. Causal Validation: Leaky Genes vs. Random Genes

A critical question is whether SHAP-identified leaky genes are causally responsible for leakage, or merely correlated. We test this by comparing the MIA reduction from perturbing the top- k leaky genes against the average MIA reduction from perturbing k randomly chosen genes (averaged over 10 random trials at noise scale $\sigma = 3.0$, with the full grid in Table V).

For the GAN generator at $k = 50$, perturbing SHAP-identified leaky genes reduces MIA AUC by **3.18 times**

more than perturbing the same number of random genes. The advantage is also positive for the GMM at $k = 50$ (2.39 $\times$) and the VAE at $k = 100$ (1.16 $\times$), confirming that the attribution identifies genuinely informative features for the regime in which each generator’s leakage is most concentrated.

At $k = 100$, the GAN advantage diminishes because the leaky set expands to include low-importance genes that add noise without additional privacy benefit. This suggests an optimal perturbation scope exists and can be identified from the SHAP concentration curve.

Negative result for VAE at $k = 50$. At $k = 50$, perturbing the VAE’s top-50 SHAP-ranked genes is less effective than random perturbation (ratio 0.06 $\times$; Table V). Two factors plausibly explain this: the VAE’s leakage is more spatially distributed than the GAN’s (lower top-10 concentration, Table II), so a 50-gene cut is too narrow to capture the leakage manifold; and the VAE’s bootstrap top-50 Jaccard is the lowest of the three generators, so the top-50 set itself is unstable. The advantage recovers at $k = 100$ once both effects are absorbed. We therefore recommend selecting k adaptively from each generator’s SHAP concentration curve rather than a fixed budget across architectures.

G. Robustness: Cross-Generator and Bootstrap Stability

Cross-generator consistency. Top-50 leaky gene sets across pairs of generators have Jaccard similarities of 0.020 (GMM vs. VAE), 0.053 (GMM vs. GAN), and 0.010 (VAE vs. GAN). These near-zero overlaps indicate that **leakage is generator-specific, not data-intrinsic**: each generator memorizes different aspects of the training data, so a leaky-gene audit performed for one architecture does not transfer to another, and the GLPA pipeline must be re-run whenever the generator changes.

Bootstrap stability. Across 10 bootstraps, individual gene rankings are unstable at the per-rank level (top-10 Jaccard 0.04 to 0.10, top-50 Jaccard 0.08 to 0.11). The exact top- k list is not reproducible from a single fit, but high-importance genes occupy a stable importance band across bootstraps, and targeted perturbation operates on sets rather than ordered lists. Combined with the causal validation in §VI-F, this is sufficient for actionable mitigation, and we recommend deployments ensemble top- k sets across bootstrap runs. Finally, a donor-scale sweep ($N \in \{20, 40, 60, 100, 200\}$) confirms these patterns persist as the cohort grows: MIA risk peaks at moderate N , concentration relaxes, and subgroup divergence sharpens, with full numbers in Appendix B.

H. Validation on Real scRNA-seq Data (PBMC)

To validate that our findings generalize beyond simulated data, we run the full GLPA pipeline on the PBMC 3k dataset [42] with 60 pseudo-donors and 978 top-variable genes. For a like-for-like comparison, we re-run the simulated pipeline with the same shadow-model count, scoring protocol, and random seed as the donor-scale sweep (Appendix B). The resulting “Sim.” rows in Table IV therefore differ by a few percentage points from the headline numbers in Sections VI-A–

TABLE IV

CROSS-DATASET COMPARISON (PBMC VS. SIMULATED, 60 DONORS).
 “SIM.” ROWS USE THE MATCHED-PROTOCOL RE-RUN DESCRIBED ABOVE.

Data	Gen.	MIA AUC	Top-50 %	Gini	Sub. Jac.	Sel. Δ	DE Pres.
PBMC	GMM	0.934	70.7	0.915	0.031	0.054	92.2
PBMC	VAE	0.712	65.1	0.898	0.099	0.116	90.7
PBMC	GAN	0.509	71.9	0.922	0.064	0.026	90.6
Sim.	GMM	0.979	65.4	0.904	0.053	0.073	91.9
Sim.	VAE	0.933	62.1	0.890	0.042	0.063	91.6
Sim.	GAN	0.667	64.2	0.898	0.075	0.012	91.3

VI-C, which use the original primary protocol. The qualitative conclusions are unchanged: high MIA AUC, Gini $\approx$ 0.9, and near-disjoint subgroup leakage.

All five core findings replicate on real data. Both informative generators leak, with PBMC MIA AUC of 0.934 for the GMM and 0.712 for the VAE. Leakage again concentrates, with Gini between 0.90 and 0.92 and the top 50 genes carrying 62 to 72% of the signal. The two subgroups leak through different gene sets, with Jaccard similarities of 0.031 to 0.099. Targeted perturbation again works: on the PBMC VAE, selective perturbation of 50 genes cuts MIA AUC from 0.712 to 0.597, the strongest selective-versus-uniform result in our experiments, while matched uniform noise achieves essentially no reduction. Finally, every PBMC perturbation preserves more than 90% of DE structure.

Why does the PBMC GAN drop to chance? Unlike the GMM and VAE, the PBMC GAN attains MIA AUC 0.509, essentially random. Two factors plausibly drive this. First, the PBMC pseudo-donor profiles, after aggregation over $\approx$ 44 cells, are much smoother than the simulated donor signatures, giving the GAN’s noise-driven generator less per-donor structure to memorize. Second, GAN training on a 30-member set is unstable. For the run we report, the discriminator collapsed to a near-uniform output before producing donor-identifiable samples. We treat the 0.509 result as a useful negative finding rather than a defense recommendation: it shows that GAN leakage is highly run- and data-dependent, which is itself an argument for per-deployment auditing rather than relying on architecture choice as a privacy safeguard. Because membership inference performance is effectively indistinguishable from chance (AUC = 0.509), the corresponding attribution, subgroup Jaccard, and perturbation results for PBMC/GAN in Table IV are not meaningfully interpretable and are therefore excluded from our quantitative analysis.

I. Pathway Analysis on Real Gene Names

As the real-data counterpart to the simulation enrichment test in §VI-D, the PBMC dataset offers a key advantage: leaky genes have real biological identities, enabling pathway enrichment analysis. We cross-reference the top-50 leaky genes against curated immune pathway gene sets (Figure 4).

We report nominal Fisher p -values (Table VIII). Across the pathways and generators tested, these do not survive Benjamini-Hochberg correction, and the contingency tables

are small (e.g., 2/4, 2/5), so the patterns below are suggestive rather than confirmatory. Three notable findings emerge:

B cell markers are enriched in GMM leaky genes (CD79A among top-50, $p = 0.014$). Since our pseudo-donor subgroups are defined by myeloid/B-cell vs. T/NK-cell enrichment, B cell marker leakage reflects how the GMM memorizes subgroup-defining features. This represents the privacy-utility tension predicted by our framework: the features that distinguish subgroups are the ones most likely to leak.

Interferon response genes are enriched across generators ($p = 0.023$ for both GMM and GAN). Interferon response genes are highly variable across individuals and conditions, making them natural memorization targets. Importantly, these genes are distinct from the primary subgroup-defining markers (T cell activation, antigen presentation), suggesting they can be perturbed with manageable biological cost for many downstream analyses.

The VAE shows no significant pathway enrichment, consistent with its leaky genes being dominated by ribosomal proteins (RPL11, RPS15, RPS23), housekeeping genes that are highly expressed and variable but not pathway-specific. This indicates that the VAE memorizes broad expression-level idiosyncrasies rather than cell-type-specific programs.

Reconciling these PBMC findings with the simulation (§VI-D), where no biological category was enriched, the two pictures are not contradictory: the simulation’s “disease genes” are an artificial 30-gene set with engineered subgroup effects, whereas real PBMC subgroups are defined by cell-type composition that is naturally encoded in lineage markers. The honest takeaway is therefore weaker than “leaky genes are biologically irrelevant”: leaky genes are not generically aligned with pathway structure, but on real data a small fraction can coincide with subgroup-defining markers (e.g., CD79A for B cells), creating a genuine privacy-utility tension that practitioners must check on their own dataset before applying targeted perturbation. Section A (“Practical deployment”) folds this check into the recommended workflow.

VII. CONCLUSION

We introduced Gene-Level Privacy Auditing (GLPA), which uses SHAP to attribute membership-inference success in synthetic scRNA-seq data to individual genes, revealing where leakage originates rather than merely whether it occurs. Across three generators, we found leakage to be concentrated in a small gene subset, divergent across subgroups, and specific to each generator. A single MIA AUC summary is therefore insufficient: audits should be feature-resolved and subgroup-stratified, which in turn enables targeted perturbation of the few responsible genes in place of blanket DP, preserving utility that uniform noise would destroy. GLPA needs only a generator that emits synthetic profiles, so natural next steps include extending it to diffusion [40] and scVI-style [38] generators, ensembling disjoint-leakage generators, and guiding per-gene noise allocation during training. We hope GLPA helps establish gene-level, subgroup-aware auditing as a routine step before any synthetic genomic release.

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APPENDIX A: EXTENDED DISCUSSION

The subgroup paradox. Near-disjoint subgroup leakage connects to the fairness-privacy literature [17], [19] but adds a dimension: disparate vulnerability manifests not just in differing attack success rates but in entirely different vulnerable feature sets, with direct implications for regulatory privacy impact assessments.

Practical deployment. We envision GLPA as a pre-release auditing step. (1) Generate synthetic data with the chosen model. (2) Audit by running GLPA to identify leaky genes per subgroup. (3) Decide: if leaky genes overlap with scientifically important features, consider more aggressive DP during training, otherwise apply targeted perturbation to the leaky subset. (4) Validate by re-running MIA on the perturbed output.

Limitations. Five caveats bound our claims. First, PBMC uses pseudo-donors rather than true biological donors, and the donor-scale sweep stops at $N=200$. Second, GLPA assumes the auditor can train shadow generators, and post-hoc perturbation is insufficient for highly memorizing models such as GMMs, which still require in-training defenses (DP-SGD, PATE). Third, SHAP rankings are unstable at the per-rank level (top-50 Jaccard 0.08 to 0.11), so deployments should ensemble top- k sets across bootstraps. Fourth, our targeted perturbation is a heuristic mitigation, not a formal (ϵ, δ) -DP

mechanism; deriving formal per-gene DP guarantees is left to future work. Fifth, our headline numbers are point estimates from a fixed $n=30$ split in a $p \gg n$ regime, so we report them as a qualitative pattern and recommend repeated-split confidence intervals before operational use.

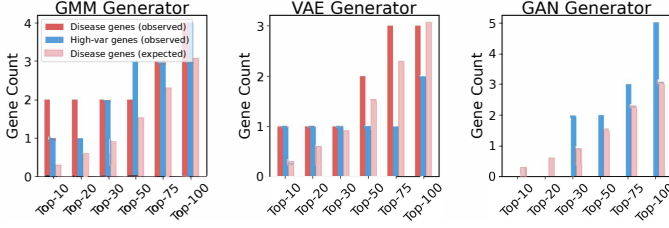


Fig. 4. Biological characterization of leaky genes. In simulation, neither disease nor high-variance genes are significantly enriched in the top-50 set; on PBMC, B-cell and interferon-response genes are enriched, exposing a privacy-utility tension.

APPENDIX B: CAUSAL VALIDATION, ABLATION, AND SCALING

Table V reports the full causal validation sweep referenced in §VI-F: for each generator and k , MIA AUC reduction from perturbing the top- k SHAP-ranked genes vs. the average reduction from 10 random k -gene selections ($\sigma = 3.0$).

TABLE V
FULL CAUSAL VALIDATION RESULTS ($\sigma = 3.0$).

Gen.	k	ΔAUC Leaky	ΔAUC Random	Ratio
GMM	50	0.024	0.010 ± 0.012	$2.39\times$
GMM	100	0.049	0.050 ± 0.026	$0.98\times$
VAE	50	0.001	0.019 ± 0.017	$0.06\times$
VAE	100	0.039	0.034 ± 0.020	$1.16\times$
GAN	25	0.049	0.034 ± 0.022	$1.44\times$
GAN	50	0.070	0.022 ± 0.033	$3.18\times$
GAN	100	0.047	0.053 ± 0.034	$0.88\times$

Figure 5 visualizes the perturbation ablation over k and noise scale σ that supports the high-noise numbers folded into §VI-E.

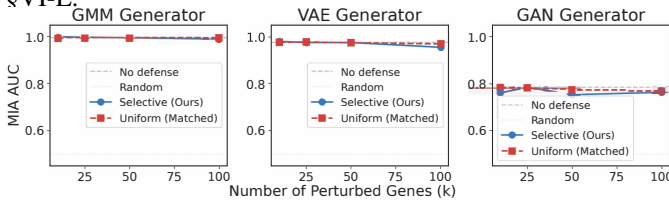


Fig. 5. Perturbation ablation over budget k and noise scale σ . Larger k and σ trade extra MIA AUC reduction for modest utility loss, and DE-preservation stays above 90% for $k \leq 50$.

Donor-scale sweep. Sweeping $N \in \{20, 40, 60, 100, 200\}$ donors on the simulated data (GMM) reveals three scaling patterns. (i) MIA vulnerability peaks at moderate N : AUC rises from 0.870 ($N=20$) to 0.979 ($N=60$), then decays to 0.930 ($N=200$) as the generator generalizes rather than memorizes. (ii) Concentration drops with more donors (top-50 share 93.6% to 48.0%, Gini 0.969 to 0.818). (iii) Subgroup divergence sharpens: Jaccard falls from 0.515 ($N=20$, noise-dominated)

to 0.020 ($N \geq 100$), confirming subgroup-specific leakage emerges only with sufficient data. Perturbation effectiveness rises in parallel, peaking at $\Delta\text{AUC} = 0.239$ ($N=200$).

Full hyperparameters, the per-method perturbation comparison, per- N numbers, and the feature-restricted MIA results are available in the released code repository.

APPENDIX C: ADDITIONAL RESULTS AND TABLES

This appendix collects the supporting tables and figures referenced from §VI-C, §VI-D, and §VI-I.

TABLE VI
SUBGROUP LEAKAGE DIVERGENCE (SIMULATED DATA). CONTROL AND DISEASE TOP-50 LEAKY-GENE SETS BARELY OVERLAP (JACCARD ≤ 0.05).

Generator	Top-50 Overlap	Jaccard Similarity
GMM	4/50	0.042
VAE	5/50	0.053
GAN	2/50	0.020

TABLE VII
BIOLOGICAL COMPOSITION OF LEAKY GENES (SIMULATED DATA). NEITHER DISEASE NOR HIGH-VARIANCE GENES ARE SIGNIFICANTLY ENRICHED IN THE TOP-50 SET (ALL $p \geq 0.46$).

Gen.	Disease Genes			High-Var Genes		
	obs/exp	Enr.	p	obs/exp	Enr.	p
GMM	2/1.5	$1.30\times$	0.461	3/2.6	$1.17\times$	0.478
VAE	2/1.5	$1.30\times$	0.461	1/2.6	$0.39\times$	0.932
GAN	0/1.5	$0.00\times$	1.000	2/2.6	$0.78\times$	0.740

TABLE VIII
PATHWAY ENRICHMENT OF LEAKY GENES (PBMC, TOP-50)

Pathway	GMM			GAN		
	obs/pres	Enr.	p	obs/pres	Enr.	p
B cell markers	2/4	$9.8\times$	0.014	0/4	$0.0\times$	1.000
IFN response	2/5	$7.8\times$	0.023	2/5	$7.8\times$	0.023
NK cytotoxicity	0/8	$0.0\times$	1.000	2/8	$4.9\times$	0.059
T cell activ.	0/9	$0.0\times$	1.000	0/9	$0.0\times$	1.000
Antigen pres.	0/12	$0.0\times$	1.000	0/12	$0.0\times$	1.000
Housekeeping	1/10	$2.0\times$	0.410	2/10	$3.9\times$	0.089